

Harsha Vardhan Polamarasetty

AI Engineer

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PROFESSIONAL SUMMARY

AI Engineer with hands-on experience developing full-stack applications, robust data pipelines, and AI-driven systems. Proficient in Python, JavaScript, and modern LLM frameworks, with practical expertise in building Retrieval-Augmented Generation (RAG) pipelines and multi-agent workflows using LangGraph, FastAPI, and leading AI models. Passionate about delivering scalable, production-ready AI solutions with clean architecture and reliable deployment practices.

SKILLS

- **AI & Machine Learning:** LangGraph, LangSmith, Retrieval-Augmented Generation (RAG), Multi-Agent Systems, Large Language Models (LLMs), BERT, Gemini API, OpenAI API
- **Programming Languages:** Python, JavaScript, TypeScript, Java, C
- **Web & API Development:** FastAPI, Flask, Django, Express.js, RESTful APIs, React, Angular, Next.js
- **Databases & DevOps:** MongoDB, PostgreSQL, MySQL, Redis, Docker, Git, CI/CD, Postman

EXPERIENCE

- **EPAM Systems**
Junior Software Engineer Jan 2026 – Jun 2026
 - Designed and built a **self-referencing database model** in MongoDB, enabling dynamic hierarchical category trees and supporting scalable product catalog structures for a production-ready eCommerce application.
 - Developed a robust full-stack application using **Angular**, **Node.js**, and **Express.js**, integrating **Redis caching** and **JWT authentication** to optimize performance and security across all user endpoints.
 - Leveraged **GitHub Copilot** to accelerate boilerplate generation and unit testing, maintaining over **80% code coverage** verified via SonarQube and Vitest within an Agile Scrum development environment.
- **Realfy Oasis**
Node.js Developer Intern Sep 2024 – Nov 2024
 - Integrated the **OpenAI API** to build an intelligent recommendation engine, processing unstructured user inputs like fitness goals and dietary preferences to dynamically generate personalized health plans.
 - Engineered secure RESTful APIs using **Node.js**, **Express.js**, and **Firebase Firestore** with **JWT authentication**, containerizing services with **Docker** for deployment on DigitalOcean cloud servers.

PROJECTS

- **RAG-based WhatsApp Chatbot for Community Platform**
LangGraph, LangSmith, FastAPI, Docker, Qdrant, OpenAI API, Google Gemini API, WhatsApp Integration, Python Aug 2025
 - Designed and implemented a **multi-agent data retrieval pipeline** using **LangGraph** to dynamically route user queries, orchestrate parallel domain-specific retrieval, and aggregate context-aware responses seamlessly.
 - Engineered a low-latency conversational pipeline integrating **Qdrant vector search** for semantic retrieval and **hierarchical rolling memory** to compress conversation history, significantly reducing token consumption.
 - Leveraged **LangSmith** for real-time agent execution tracing, systematic debugging, and prompt optimization to maintain consistently high-quality response generation, system reliability, and agent performance.
- **AI-Powered News Analyzer News Analysis & Summarization**
Python, Flask, Gradio, BeautifulSoup, BERT, Gemini API, gTTS Mar 2024
 - Engineered an automated news analysis pipeline leveraging **BeautifulSoup** for web scraping and **BERT** with **Gemini API** to extract unstructured data, generating accurate executive summaries efficiently.
 - Developed an interactive full-stack interface using **Flask** and **Gradio** to present real-time sentiment trends, key topic insights, and interactive audio narration generated via **gTTS integration**.
 - Optimized data processing workflows and text extraction routines to transform raw web articles into structured, actionable insights with minimal latency and high parsing accuracy across varied sources.
- **MusiBuzz Music Streaming & Recommendation Platform**
Python, Django, Vue.js, Tailwind CSS, RESTful APIs, Spotify API, Collaborative Filtering Nov 2023
 - Engineered a custom **collaborative filtering recommendation system** analyzing user behavior vectors to identify shared listening patterns and dynamically serve highly tailored track recommendations.
 - Developed a full-stack music discovery web application with a **Django** backend and **Vue.js** frontend, optimizing real-time external database search queries and overall user interface responsiveness.

EDUCATION

- **KL University**
B.Tech in Computer Science; 9.5 / 10 Aug 2022 – Apr 2026